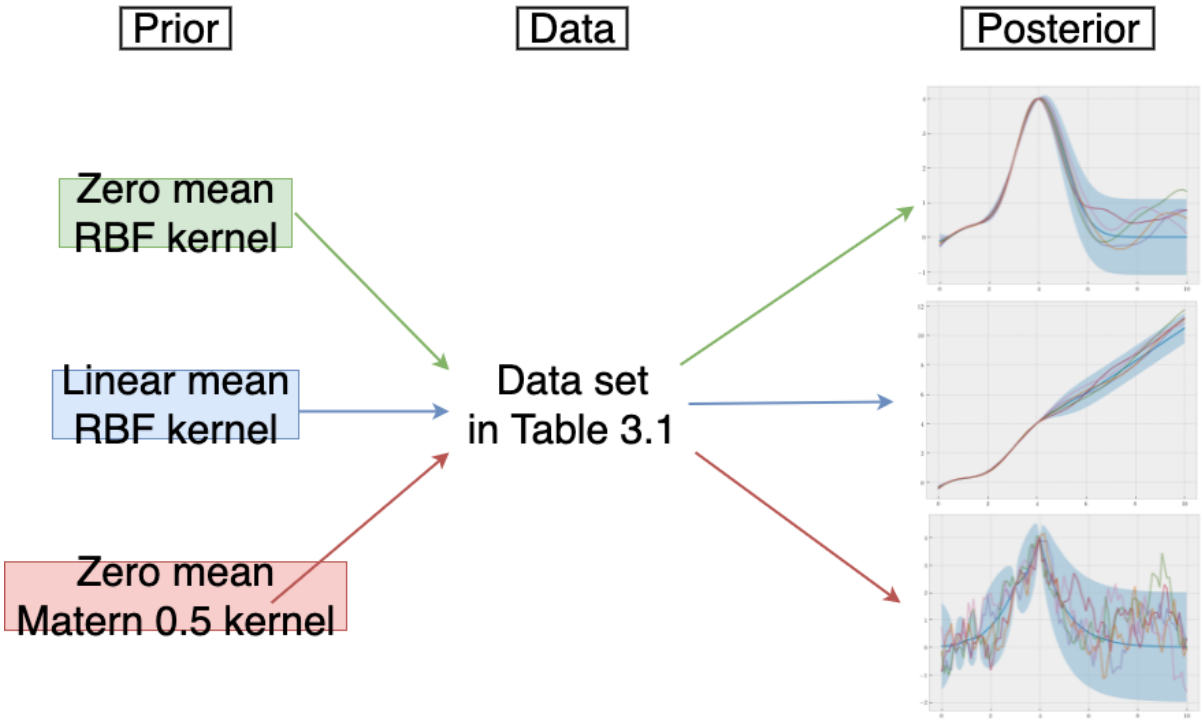


Figure 3.15 Three different choices for the mean function and kernel and the predictions made by their respective posterior GPs when trained on the same data set. Each choice leads to a different behavior in prediction.

3.4.3 Modeling different levels of variability with multiple length scales

As we have only been considering one-dimensional objective functions, there is only one length scale that we need to consider. However, we can imagine scenarios in which a high-dimensional objective function has more variability in some dimensions and is smoother in other dimensions. That is, some dimensions have small length scales while others have large length scales. Remember our motivating example at the beginning of this chapter: the price prediction of a house increases by a larger amount with an extra story than with an extra squared foot in living area. We will see how we can maintain multiple length scales in a GP to model these functions in this subsection.

If we were to only use a single length scale for all dimensions, we wouldn't be able to faithfully model the objective function. This situation calls for the GP model to maintain a separate length scale for each dimension, in order to fully capture the variability in each of them. In this and final section of the chapter, we will learn how to do this with GPyTorch.

To aid our discussion, we will be using a concrete two-dimensional objective function called Ackley, which can be modified to have various levels of variability in different dimensions, as we will see shortly. We implemente the function as follows:

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